Method of Pole Expansions, Series Expansions of Jacobi Elliptic Function, and Properties of Dixon Elliptic Functions

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1 Abstract

In this paper, we will study how to give pole expansions of certain functions by providing a formula for the coefficients of the pole expansion of a function. We will then apply this formula to certain functions, in particular, the Jacobi elliptic functions. In doing so, we will see that a formula for the number of ways a number can be written as a sum of 4 odd squares will be given as a corollary. We will then apply this method to Dixon elliptic functions to find various infinite series and product representations and investigate several of their properties.

2 A Formula for Pole Expansions of Certain Meromorphic Functions

Suppose we have a function $f(z)$ that has n distinct poles $p_1, p_2, \ldots, p_n$ and these poles have order $j(p_1), j(p_2), \ldots, j(p_n)$ respectively. That is, given a pole p, $j(p)$ is its order. Then, we wish to expand $f(z)$ as

$$f(z) = C + \sum_{m=1}^{n} \sum_{k=1}^{j(p_m)} \frac{A_k(p_m)}{(z-p_m)^k}$$

Writing out the first few terms, we see that $f(z) = \frac{A_1(p_1)}{z-p_1} + \frac{A_2(p_1)}{(z-p_1)^2} + ... + \frac{A_{j(p_1)}(p_1)}{(z-p_1)^{j(p_1)}} + g(z)$, where $g(z)$ has no poles at $z = p_1$. From this formulation, we can see that

$$A_k(p_1) = \frac{1}{(j(p_1)-k)!} \lim_{z \to 1} \frac{d^{j(p_1)-k}}{dz^{j(p_1)-k}} (z-p_1)^{j(p_1)} f(z).$$

This formula holds for all the other poles as well. Thus, we have the following theorem:

Theorem 1. *Let $f(z)$ be a function with poles at $z = p_1, p_2, \ldots, p_n$ each with order $j(p_1), j(p_2), \ldots, j(p_n)$ respectively. Under certain growth restrictions on $f(z)$, $f(z)$ may be expanded as*

$$f(z) = C + \sum_{m=1}^{n} \sum_{k=1}^{j(p_m)} \frac{A_k(p_m)}{(z-p_m)^k},$$

where $A_k(p_m) = \frac{1}{(j(p_m)-k)!} \lim_{z\to\ p_m} \frac{d^{j(p_m)-k}}{dz^{j(p_m)-k}} (z-p_m)^{j(p_m)} f(z)$ *for* $m = 1, 2, \ldots, n$.

As a preliminary example, we will apply the method to $f(z) = \frac{1}{\sinh^2(\pi z)}$. This function has poles at $z = 0, \pm i, \pm 2i, \pm 3i$, and so on. Furthermore, all of these poles are of order 2, so $j(p_m) = 2$ for all poles p_m. Thus,

$$\frac{1}{\sinh^2(\pi z)} = C + \sum_{m=-\infty}^{\infty} \left(\frac{A_1(im)}{z-im} + \frac{A_2(im)}{(z-im)^2} \right).$$

Since $f(z)$ is an even function, it can be shown that $A_1(im) = 0$ for all integers m. By our formula for the coefficients of the pole expansion, we have that $A_2(im) = \lim_{z\to} \frac{(z-im)^2}{\sinh^2(\pi z)} = \frac{1}{\pi^2}$. From this, we have that

$$\frac{1}{\sinh^2(\pi z)} - \frac{1}{(\pi z)^2} = C + \frac{1}{\pi^2} \left(\sum_{m=1}^{\infty} \frac{1}{(z-im)^2} + \frac{1}{(z+im)^2} \right).$$

Taking the limit as $z \to 0$, we have that $C = 0$. Thus,

$$\frac{1}{\sinh^2(\pi z)} = \frac{1}{\pi^2} \sum_{m=-\infty}^{\infty} \frac{1}{(z-im)^2} = \frac{1}{\pi^2 z^2} + \frac{2}{\pi^2} \sum_{m=1}^{\infty} \frac{z^2 - n^2}{(z^2+n^2)^2}.$$

Similarly, it can be shown that

$$\frac{1}{\sinh^3(\pi z)} = \frac{1}{\pi^3 z^3} - \frac{1}{2\pi z} + \frac{2}{\pi} \sum_{m=1}^{\infty} \frac{(-1)^{m-1}}{z^2+m^2} + \frac{2z}{\pi^3} \sum_{m=1}^{\infty} \frac{(-1)^m (z^2 - 3m^2)}{(z^2+m^2)^3}.$$

3 Series Expansions of Jacobi Elliptic Functions and Sum of 4 Odd Squares

We will now apply the method of pole expansions to the Jacobi elliptic functions $\mathrm{sn}(z)$, $\mathrm{cn}(z)$, $\mathrm{dn}(z)$. The function $\mathrm{sn}(z)$ has only simple poles at $z = 2mK + (2n+1)iK'$ for m and n being integers. Thus, it may be expanded as

$$\mathrm{sn}(z) = C + \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \frac{A_1(2mK + (2n+1)iK')}{z - 2mK - (2n+1)iK'}$$

. Furthermore, $\mathrm{sn}(z)$ is an odd function, so it can be shown that $C = 0$. We also have that $A_1(2mK + (2n+1)iK')$ is simply the residue of $\mathrm{sn}(z)$ at

$z = 2mK + (2n+1)\,iK'$, which is $\frac{(-1)^m}{k}$. Thus, we have the following series expansion:

$$\operatorname{sn}(z) = \frac{1}{k}\sum_{m=-\infty}^{\infty}\sum_{n=-\infty}^{\infty}\frac{(-1)^m}{z - 2mK - (2n+1)\,iK'}.$$

Using the pole expansion formula for the hyperbolic tangent, we get that

1. $$\operatorname{sn}(z) = \frac{\pi}{2kK'}\sum_{m=-\infty}^{\infty}(-1)^m \tanh\left(\frac{\pi}{2K'}(z - 2mK)\right).$$

We can use the same procedure for the other two elliptic functions to get that

1. $$\operatorname{sn}(2Kz) = \frac{\pi}{2kK'}\sum_{m=-\infty}^{\infty}(-1)^m \tanh\left(\frac{\pi K}{K'}(z-m)\right)$$

$$= \frac{2\pi\sin(\pi z)}{kK}\sum_{n=0}^{\infty}\frac{\cosh\left(\frac{(2n+1)\pi}{2}\frac{K'}{K}\right)}{\cosh\left((2n+1)\pi\frac{K'}{K}\right) - \cos(2\pi z)}$$

2. $$\operatorname{cn}(2Kz) = \frac{\pi}{2kK'}\sum_{m=-\infty}^{\infty}(-1)^m \operatorname{sech}\left(\frac{\pi K}{K'}(z-m)\right)$$

$$= \frac{2\pi\cos(\pi z)}{kK}\sum_{n=0}^{\infty}\frac{(-1)^n\sinh\left(\frac{(2n+1)\pi}{2}\frac{K'}{K}\right)}{\cosh\left((2n+1)\pi\frac{K'}{K}\right) - \cos(2\pi z)}$$

3. $$\operatorname{dn}(2Kz) = \frac{\pi}{2K'}\sum_{m=-\infty}^{\infty}\operatorname{sech}\left(\frac{\pi K}{K'}(z-m)\right)$$

$$= \frac{\pi\sec^2(\pi z)}{2K}\sum_{n=-\infty}^{\infty}\frac{(-1)^n\tanh\left(\frac{(2n+1)\pi}{2}\frac{K'}{K}\right)}{\tanh^2\left(\frac{(2n+1)\pi}{2}\frac{K'}{K}\right) + \tan^2(\pi z)}$$

4. $$\operatorname{sn}^2(2Kz) = \frac{\pi^2}{4k^2K'^2}\sum_{m=-\infty}^{\infty}\left(\operatorname{sech}^2\left(\frac{\pi mK}{K'}\right) - \operatorname{sech}^2\left(\frac{\pi K}{K'}(z-m)\right)\right)$$

$$= \frac{\pi^2}{4k^2K^2}\sum_{n=0}^{\infty}\left(\frac{2}{\cosh^2\left(\frac{(2n+1)\pi}{2}\frac{K'}{K}\right)} + \frac{\cos(2\pi z)\cosh\left((2n+1)\pi\frac{K'}{K}\right) - 1}{\left(\cosh\left((2n+1)\pi\frac{K'}{K}\right) - \cos(2\pi z)\right)^2}\right)$$

5.

$$\mathrm{cn}^2(2Kz) = 1 + \frac{\pi^2}{4k^2K'^2} \sum_{m=-\infty}^{\infty} \left(\mathrm{sech}^2 \left(\frac{\pi K}{K'} (z-m) \right) - \mathrm{sech}^2 \left(\frac{\pi m K}{K'} \right) \right)$$

$$1 + \frac{\pi^2}{4k^2K^2} \sum_{n=0}^{\infty} \left(\frac{1 - \cos(2\pi z) \cosh\left((2n+1)\pi \frac{K'}{K}\right)}{\left(\cosh\left((2n+1)\pi \frac{K'}{K}\right) - \cos(2\pi z)\right)^2} - \frac{2}{\cosh^2\left(\frac{(2n+1)\pi}{2} \frac{K'}{K}\right)} \right)$$

6.

$$\mathrm{dn}^2(2Kz) = 1 + \frac{\pi^2}{4K'^2} \sum_{m=-\infty}^{\infty} \left(\mathrm{sech}^2 \left(\frac{\pi K}{K'} (z-m) \right) - \mathrm{sech}^2 \left(\frac{\pi m K}{K'} \right) \right)$$

Some simple identities that follow from the series above are

1.

$$\left[\sum_{m=-\infty}^{\infty} \mathrm{sech} \left(\frac{\pi K}{K'} (z+m) \right) \right]^2 - \left[\sum_{m=-\infty}^{\infty} (-1)^m \mathrm{sech} \left(\frac{\pi K}{K'} (z+m) \right) \right]^2 = \left(\frac{2kK}{\pi} \right)^2$$

2.

$$\sum_{n=-\infty}^{\infty} \frac{\cosh\left((2n+1)\pi \frac{K'}{K}\right)}{\sinh^2\left((2n+1)\pi \frac{K'}{K}\right)} = \left(\frac{kK}{\pi} \right)^2$$

Due to the K-formula for $\vartheta_2(q)$, identity (2) actually implies the following number theoretic result:

3.

$$16 \sum_{n=0}^{\infty} a(2n+1) q^{4(2n+1)} = \sum_{n=0}^{\infty} s_4(n) q^n,$$

where $s_4(n)$ is the number of ways n can be expressed as a sum of 4 odd squares and $a(n)$ is the sum of all odd divisors of n. This implies the following formula for $s_4(n)$:

$$s_4(n) = \begin{cases} 16a\left(\frac{n}{4}\right) & \text{If n is an odd multiple of 4} \\ 0 & \text{otherwise} \end{cases}$$

In particular, if p is an odd prime number, then $s_4(4p) = 16(p+1)$.

4 Properties of Dixon Elliptic Functions

The Dixon elliptic functions are a set of elliptic functions denoted by $\mathrm{sm}(z)$ and $\mathrm{cm}(z)$. These functions are relatively unknown when compared to their Jacobian counterpart, and in this section, I will give several properties satisfied by these functions, specifically series and product representations, and also study a sequence of numbers related to these functions as well as the Eisenstein integers.

To start off, I will mention some known fundamental properties of the Dixon elliptic functions. The Dixon elliptic functions are expressed as the inverses to the following elliptic integrals

$$z = \int_0^{\mathrm{sm}(z)} \frac{1}{(1-t^3)^{\frac{2}{3}}} dt = \int_{\mathrm{cm}(z)}^1 \frac{1}{(1-t^3)^{\frac{2}{3}}} dt$$

From these inverse relations it can be seen that thee functions have a Pythagorean-like identity:

$$\mathrm{sm}^3(z) + \mathrm{cm}^3(z) = 1.$$

Furthermore, let $\pi_3 = \frac{\sqrt{3}\Gamma^3\left(\frac{1}{3}\right)}{2\pi}$ and $\omega = e^{\frac{2\pi i}{3}}$. Then, these functions have fundamental periods π_3 and $\omega\pi_3$. Thus, all periods of these functions are given by the Eisenstein integers scaled by the value π_3. Also, $\mathrm{sm}\,(z) = \mathrm{cm}\left(\frac{\pi_3}{3} - z\right)$. Let a and b be integers. Then, the function $\mathrm{sm}(z)$ has its zeros and poles at $\frac{i\pi_3}{\sqrt{3}}(a+b\omega)$ and $-\frac{\pi_3}{3} + \frac{i\pi_3}{\sqrt{3}}(a+b\omega)$ respectively. The function $\mathrm{cm}(z)$ has its zeros and poles at $\frac{\pi_3}{3} + \frac{i\pi_3}{\sqrt{3}}(a+b\omega)$ and $-\frac{\pi_3}{3} + \frac{i\pi_3}{\sqrt{3}}(a+b\omega)$ respectively.

Knowing these facts about the Dixon elliptic functions and using the pole expansion formula, we can find the following series representations for the Dixon elliptic functions:

1.
$$\frac{1}{\mathrm{sm}(\pi_3 z)} = \mathrm{sm}\left(-\pi_3 z - \frac{\pi_3}{3}\right)$$

$$= \frac{1}{2} + \frac{\pi}{\pi_3}\sum_{n=0}^{\infty}\left(\frac{(-1)^n\cos\left(\frac{2\pi n}{3}\right)\sin(2\pi z) + \sin\left(\frac{2\pi n}{3}\right)\sinh\left(\frac{\pi n}{\sqrt{3}}\right)}{\cosh\left(\frac{\pi n}{\sqrt{3}}\right) + (-1)^{n+1}\cos(2\pi z)} - \frac{2\sin\left(\frac{2\pi n}{3}\right)\sinh\left(\frac{\pi n}{\sqrt{3}}\right)}{2\cosh\left(\frac{\pi n}{\sqrt{3}}\right) + (-1)^n}\right)$$

2.
$$\frac{\pi_3}{2\pi}\left[\frac{1}{\mathrm{sm}(\pi_3 z)} + \frac{1}{\mathrm{sm}(-\pi_3 z)} + 1\right] = \sum_{n=-\infty}^{\infty}\left(\frac{\sin\left(\frac{2\pi n}{3}\right)\sinh\left(\frac{\pi n}{\sqrt{3}}\right)}{\cosh\left(\frac{\pi n}{\sqrt{3}}\right) + (-1)^{n+1}\cos(2\pi z)} - \frac{2\sin\left(\frac{2\pi n}{3}\right)\sinh\left(\frac{\pi n}{\sqrt{3}}\right)}{2\cosh\left(\frac{\pi n}{\sqrt{3}}\right) + (-1)^n}\right)$$

3.
$$\frac{\pi_3}{2\pi}\left[\frac{1}{\mathrm{sm}(\pi_3 z)} - \frac{1}{\mathrm{sm}(-\pi_3 z)}\right] = \frac{\sin(2\pi z)}{2}\sum_{n=-\infty}^{\infty}\frac{\cos\left(\frac{2\pi n}{3}\right)}{(-1)^n\cosh\left(\frac{\pi n}{\sqrt{3}}\right) - \cos(2\pi z)}$$

4.

$$\frac{1}{\mathrm{sm}^2(\pi_3 z)\mathrm{sm}^2(-\pi_3 z)} = \frac{12\pi^4}{\pi_3^4}\sum_{n=-\infty}^{\infty}\frac{2+(-1)^n\cosh\left(\sqrt{3}\pi\left(z+n\right)\right)}{\left((-1)^n\cosh\left(\sqrt{3}\pi\left(z+n\right)\right)-1\right)^2}$$

5.

$$\frac{1}{\mathrm{sm}^3\left(\frac{i\pi_3 z}{2\sqrt{3}}\right)} = \frac{1}{2}+\frac{i\pi^3}{\pi_3^3}\sum_{n=-\infty}^{\infty}\frac{(-1)^n\left(3+\cosh\left(\frac{2\pi}{\sqrt{3}}\left(z+n\right)\right)\right)+4\cosh\left(\frac{\pi}{\sqrt{3}}\left(z+n\right)\right)}{\sinh^3\left(\frac{\pi}{\sqrt{3}}\left(z+n\right)\right)}$$

6.

$$\frac{\pi_3^2}{\mathrm{sm}(\pi_3 z)\mathrm{sm}(-\pi_3 z)} = 2\sqrt{3}+2\pi^2\sum_{n=-\infty}^{\infty}\frac{(-1)^n\cos\left(2\pi z\right)\cosh\left(\frac{\pi n}{\sqrt{3}}\right)-1}{\left((-1)^n\cosh\left(\frac{\pi n}{\sqrt{3}}\right)-\cos\left(2\pi z\right)\right)^2}$$

From which it follows that

$$\sqrt[3]{2} = \frac{1}{2}+\frac{\pi}{\pi_3}\sum_{n=0}^{\infty}\left(\frac{(-1)^n\cos\left(\frac{2\pi n}{3}\right)+\sin\left(\frac{2\pi n}{3}\right)\sinh\left(\frac{\pi n}{\sqrt{3}}\right)}{2\cosh\left(\frac{\pi n}{\sqrt{3}}\right)+(-1)^{n+1}\sqrt{3}}-\frac{2\sin\left(\frac{2\pi n}{3}\right)\sinh\left(\frac{\pi n}{\sqrt{3}}\right)}{2\cosh\left(\frac{\pi n}{\sqrt{3}}\right)+(-1)^n}\right)$$

$$\sqrt[6]{2}\sqrt[4]{3} = \frac{2\pi}{\pi_3}\sum_{n=-\infty}^{\infty}\frac{(-1)^n\cos\left(\frac{2\pi n}{3}\right)}{\cosh\left(\frac{\pi n}{\sqrt{3}}\right)}$$

$$\frac{\sqrt[3]{2}}{\sqrt{3}} = \frac{4\pi}{\pi_3}\sum_{n=-\infty}^{\infty}\frac{\cos\left(\frac{2\pi n}{3}\right)}{2\cosh\left(\frac{\pi n}{\sqrt{3}}\right)+1}$$

$$\sum_{n=0}^{3}(-1)^n\,\mathrm{sm}\left(\frac{(6n+1)\,\pi_3}{24}\right) = \frac{4\pi}{\pi_3}\sum_{n=-\infty}^{\infty}\frac{\cos\left(\frac{2\pi n}{3}\right)}{\cosh\left(\frac{2\pi n}{\sqrt{3}}\right)}$$

The last sum, much like the previous sums, is also algebraic. This follows since elliptic functions have algebraic addition formulas. Let $\alpha = \frac{-1+\sqrt{3+2\sqrt{3}}}{\sqrt[3]{4}}$ and $\beta = \frac{-1-\sqrt{3+2\sqrt{3}}}{\sqrt[3]{4}}$. Let x_1 and x_2 be the real roots of $\alpha x^4 - 2x^3 + 2\alpha x - 1$ and x_3 and x_4 be the real roots of $\beta x^4 + 2x^3 - 2\beta x - 1$. Then,

$$\frac{4\pi}{\pi_3}\sum_{n=-\infty}^{\infty}\frac{\cos\left(\frac{2\pi n}{3}\right)}{\cosh\left(\frac{2\pi n}{\sqrt{3}}\right)} = x_1+x_2+x_3+x_4,$$

which shows that the series above is indeed algebraic.

Using these series representations for the Dixon functions, we can find Fourier

series representations of these functions. From formula (3) in this section, we have that

$$\frac{\pi_3}{2\pi}\left[\frac{1}{\mathrm{sm}(\pi_3 z)}-\frac{1}{\mathrm{sm}(-\pi_3 z)}\right]=\frac{\sin(2\pi z)}{2}\sum_{n=-\infty}^{\infty}\frac{\cos\left(\frac{2\pi n}{3}\right)}{(-1)^n\cosh\left(\frac{\pi n}{\sqrt{3}}\right)-\cos(2\pi z)}$$

The summand is itself a periodic function, so we can find its Fourier series expansion. Doing so, we get

$$\frac{\cos\left(\frac{2\pi n}{3}\right)}{(-1)^n\cosh\left(\frac{\pi n}{\sqrt{3}}\right)-\cos(2\pi z)}=\frac{(-1)^n\cos\left(\frac{2\pi n}{3}\right)}{\sinh\left(\frac{\pi n}{\sqrt{3}}\right)}\left(1+2\sum_{m=1}^{\infty}(-1)^{mn}e^{-\frac{\pi mn}{\sqrt{3}}}\cos(2\pi mz)\right).$$

Summing both sides, we have

$$\sum_{n=1}^{\infty}\frac{\cos\left(\frac{2\pi n}{3}\right)}{(-1)^n\cosh\left(\frac{\pi n}{\sqrt{3}}\right)-\cos(2\pi z)}$$

$$=\sum_{n=1}^{\infty}\frac{(-1)^n\cos\left(\frac{2\pi n}{3}\right)}{\sinh\left(\frac{\pi n}{\sqrt{3}}\right)}+\sum_{m=1}^{\infty}\left(\sum_{n=1}^{\infty}\frac{2(-1)^{(m+1)n}e^{-\frac{\pi mn}{\sqrt{3}}}\cos\left(\frac{2\pi n}{3}\right)}{\sinh\left(\frac{\pi n}{\sqrt{3}}\right)}\right)\cos(2\pi mz)$$

Using the fact that $\sin(2\pi z)\cos(2\pi mz)=\frac{1}{2}\left(\sin(2\pi(m+1)z)-\sin(2\pi(m-1)z)\right)$ and appropriately reindexing the sum, we get the following Fourier series:

$$\frac{\pi_3}{2\pi}\left[\frac{1}{\mathrm{sm}(\pi_3 z)}-\frac{1}{\mathrm{sm}(-\pi_3 z)}\right]=\frac{\sin(2\pi z)}{2}\sum_{n=-\infty}^{\infty}\frac{\cos\left(\frac{2\pi n}{3}\right)}{(-1)^n\cosh\left(\frac{\pi n}{\sqrt{3}}\right)-\cos(2\pi z)}$$

$$=\cot(\pi z)-2\sum_{m=1}^{\infty}\frac{(-1)^m e^{-\frac{\pi m}{\sqrt{3}}}\left(2(-1)^m e^{-\frac{\pi m}{\sqrt{3}}}+1\right)}{e^{-\frac{2\pi m}{\sqrt{3}}}+(-1)^m e^{-\frac{\pi m}{\sqrt{3}}}+1}\sin(2\pi mz).$$

We can use this method on the other series expansions to find more Fourier series. From here on out, unless stated otherwise, let $q=-e^{-\frac{\pi}{\sqrt{3}}}$. Then, we have the following Fourier series:

7.

$$\frac{\pi_3}{2\pi}\left[\frac{1}{\mathrm{sm}(\pi_3 z)}-\frac{1}{\mathrm{sm}(-\pi_3 z)}\right]=\cot(\pi z)-2\sum_{m=1}^{\infty}\frac{q^m(2q^m+1)}{1+q^m+q^{2m}}\sin(2\pi mz)$$

8.

$$\frac{\pi_3}{2\pi}\left[\frac{1}{\mathrm{sm}(\pi_3 z)}+\frac{1}{\mathrm{sm}(-\pi_3 z)}\right]=\sqrt{3}\sum_{m=-\infty}^{\infty}\frac{q^m}{1+q^m+q^{2m}}\cos(2\pi mz)$$

9.

$$\frac{\pi_3}{\pi}\frac{1}{\mathrm{sm}(\pi_3 z)} = \frac{1}{\sqrt{3}} + \cot(\pi z) + 2\sum_{m=1}^{\infty} \frac{q^m\left(\sqrt{3}\cos(2\pi m z) - (2q^m+1)\sin(2\pi m z)\right)}{1+q^m+q^{2m}}$$

10.

$$\frac{\pi_3^2}{\mathrm{sm}(\pi_3 z)\mathrm{sm}(-\pi_3 z)} = 2\sqrt{3}\pi - \frac{\pi^2}{\sin^2(\pi z)} + 8\pi^2 \sum_{m=1}^{\infty} \frac{m q^m}{1-q^m}\cos(2\pi m z)$$

11.

$$\log\left|\mathrm{sm}\left(\frac{\pi_3 z}{2}\right)\right| = \log\left|\frac{\sin\left(\frac{\pi x}{2}\right)}{\sin\left(\frac{\pi x}{2}+\frac{\pi}{3}\right)}\right| + 4\sum_{m=1}^{\infty}\frac{q^m \sin\left(\frac{\pi m}{3}\right)}{m(1-q^m)}\sin\left(\pi m\left(z+\frac{1}{3}\right)\right)$$

Using the fact $\sin\left(\pi m\left(x+\frac{1}{3}\right)\right)\sin\left(\frac{\pi m}{3}\right) = \frac{\cos(\pi m x)-\cos\left(\pi m x+\frac{2\pi m}{3}\right)}{2}$, we can expand $\mathrm{sm}(z)$ as an infinite product:

12.

$$\mathrm{sm}\left(\frac{\pi_3 z}{\pi}\right) = \frac{\sin(z)}{\sin\left(z+\frac{\pi}{3}\right)}\prod_{n=1}^{\infty}\frac{1-2q^n\cos(2z)+q^{2n}}{1-2q^n\cos\left(2z+\frac{2\pi}{3}\right)+q^{2n}}$$

13.

$$\mathrm{cm}\left(\frac{\pi_3 z}{\pi}\right) = \frac{\sin\left(\frac{\pi}{3}-z\right)}{\sin\left(\frac{\pi}{3}+z\right)}\prod_{n=1}^{\infty}\frac{1-2q^n\cos\left(\frac{2\pi}{3}-2z\right)+q^{2n}}{1-2q^n\cos\left(\frac{2\pi}{3}+2z\right)+q^{2n}}$$

The known identity that $\mathrm{cm}(z)\mathrm{cm}(-z) = 1$ can be seen very clearly from the infinite product of $\mathrm{cm}(z)$. Thus, both functions can be written as a ratio of entire functions, which are related to the Jacobi theta functions:

14.

$$\mathrm{sm}(z) = \frac{M(z)}{M\left(z+\frac{\pi_3}{3}\right)}$$

15.

$$\mathrm{cm}(z) = \frac{M\left(\frac{\pi_3}{3}-z\right)}{M\left(\frac{\pi_3}{3}+z\right)},$$

where $M(z) = 2\sin\left(\frac{\pi z}{\pi_3}\right)\prod_{n=1}^{\infty}\left(1-q^n\cos\left(\frac{2\pi z}{\pi_3}\right)+q^{2n}\right)$ is entire. It can be seen that much like the case of the Lemniscate elliptic functions, the logarithm of $M(z)$ takes simple forms for certain values. For example,

16.

$$\log M\left(\frac{\pi_3}{6}\right) = \log\left(\frac{M\left(\frac{\pi_3}{6}\right)}{\sqrt[4]{3}}\right) = \frac{\pi}{12\sqrt{3}}.$$

The proof of (16) is quite simple to see when using the theory of elliptic integrals. We have that

$$M\left(\frac{\pi_3}{6}\right)=\prod_{n=1}^{\infty}\left(1-(-1)^n e^{-\frac{\pi n}{\sqrt{3}}}+e^{-\frac{2\pi n}{\sqrt{3}}}\right)=\prod_{n=1}^{\infty}\frac{1+(-1)^n e^{-\pi n\sqrt{3}}}{1+(-1)^n e^{-\frac{\pi n}{\sqrt{3}}}}.$$

Consider the following product for arbitrary q:

$$\prod_{n=1}^{\infty}\left(1+(-1)^n q^n\right)=\prod_{n=1}^{\infty}\left(1-q^{2n-1}\right)\left(1+q^{2n}\right)$$

It is known that when $q=\exp\left(-\pi\frac{K'}{K}\right)$, then

$$\prod_{n=1}^{\infty}\left(1+q^{2n}\right)=2^{-\frac{1}{3}}q^{-\frac{1}{12}}k^{\frac{1}{6}}k'^{-\frac{1}{12}}$$

$$\prod_{n=1}^{\infty}\left(1+q^{2n}\right)=2^{-\frac{1}{3}}q^{-\frac{1}{12}}k^{\frac{1}{6}}k'^{-\frac{1}{12}}.$$

Thus,

$$\prod_{n=1}^{\infty}\left(1+(-1)^n e^{-\pi n\frac{K'}{K}}\right)=2^{-\frac{1}{6}}e^{\frac{\pi}{24}\frac{K'}{K}}\left(kk'\right)^{\frac{1}{12}}.$$

Switching k and k' gives

$$\prod_{n=1}^{\infty}\left(1+(-1)^n e^{-\pi n\frac{K}{K'}}\right)=2^{-\frac{1}{6}}e^{\frac{\pi}{24}\frac{K}{K'}}\left(kk'\right)^{\frac{1}{12}}$$

. Dividing these two products and letting $\frac{K'}{K}=\sqrt{m}$, we obtain

$$\prod_{n=1}^{\infty}\frac{1+(-1)^n e^{-\pi n\sqrt{m}}}{1+(-1)^n e^{-\frac{\pi n}{\sqrt{m}}}}=e^{\frac{\pi}{24}\left(\sqrt{m}-\frac{1}{\sqrt{m}}\right)}.$$

Evaluating the product at $m=3$ gives the desired result. The evaluation of $M\left(\frac{\pi_3}{3}\right)$ follows similarly, but we will instead have to use the identities $\prod_{n=1}^{\infty}\left(1+q^{2n-1}\right)=2^{\frac{1}{6}}q^{\frac{1}{24}}\left(kk'\right)^{-\frac{1}{12}}$ and $\prod_{n=1}^{\infty}\left(1-q^{2n}\right)=2^{\frac{1}{6}}q^{-\frac{1}{12}}\left(kk'\right)^{\frac{1}{6}}\sqrt{\frac{K}{\pi}}$.

We will now look at a new sequence of numbers R_n, which are intimately related to the Dixon elliptic functions and the sums of reciprocals of Eisenstein integers (i.e., the Eisenstein series evaluated at $\omega=\exp\frac{2i\pi}{3}$).

Consider the Eisenstein series $G_{2k}(\omega)$:

$$G_{2k}(\omega)=\sum_{(m,n)\in\mathbb{Z}^2/(0,0)}\frac{1}{(m+n\omega)^{2k}},$$

where k is a positive integer. If $k \not\equiv 0 \mod 3$ and $k \geq 2$, then $G_{2k}(\omega) = 0$. If k is divisible by 3, then let R_{2k} be a sequence of numbers such that $G_{2k}(\omega) = \frac{R_{2k}\pi_3^{2k}}{(2k)!}$. If we let $R_k = 0$ for k not divisible by 6, then $G_k(\omega) = \frac{R_k \pi_3^k}{k!}$ for $k \geq 3$. By finding the pole expansions of various combinations of the Dixon elliptic functions and then re-expressing the pole expansions as power series, it can be seen that the sequence of numbers R_{6n} can be used to express the power series of the Dixon elliptic functions. For example,

17.
$$\frac{\mathrm{cm}(z)}{\mathrm{sm}^2(z)} = \frac{1}{z^2} + \sum_{n=1}^{\infty} \frac{(-1)^n 3^{3n} R_{6n}}{6n} \frac{z^{6n-2}}{(6n-2)!}$$

for $|z| < \frac{\pi_3}{\sqrt{3}}$. This series is actually the elliptic and Eisenstein integer analog to the series $\frac{1}{\sin^2(z)} = \frac{1}{z^2} - \sum_{n=1}^{\infty} \frac{(-1)^n 2^{2n} B_{2n}}{2n} \frac{z^{2n-2}}{(2n-2)!}$. Therefore, whereas the Bernoulli numbers are used to express power series of the usual trigonometric functions, which can be thought of as degenerate elliptic functions, whose period lattice is just the integers scaled by 2π (which can be thought of as a degenerate one-dimensional lattice), the sequence R_{6n} is the equivalent for a lattice comprised of the Eisenstein integers scaled by π_3.

Another power series given by R_{6n} is

18.
$$\frac{1}{\mathrm{sm}^3(z)} = \frac{1}{z^3} + \frac{1}{2} + \sum_{n=1}^{\infty} \frac{(-1)^{n-1} 3^{3n} R_{6n}}{12n} \frac{z^{6n-3}}{(6n-3)!}$$

for $|z| < \frac{\pi_3}{\sqrt{3}}$. The function $y = \frac{\mathrm{cm}(z)}{\mathrm{sm}^2(z)}$ satisfies the differential equation $y'' = 6y^2$. This implies the following recursive formula for R_{6n}:

19.
$$R_{6n} = \frac{n}{(6n+1)(n-1)} \sum_{k=1}^{n-1} \binom{6n-2}{6k-2} \frac{6n-6k-1}{k} R_{6k} R_{6(n-k)}$$

for $n > 1$. Since this formula expresses R_{6n} as rational combinations of R_{6m} for $m < n$, R_{6n} is rational for all $n \geq 1$ if and only if R_6 is rational. It may be shown that $R_6 = \frac{4}{21}$, so R_{6n} is always rational. Using this recursive formula, we find the first few values of R_{6n} to be $R_6 = \frac{4}{21}, R_{12} = \frac{1600}{273}, R_{18} = \frac{1408000}{399}, R_{24} = \frac{4212736000}{273}, R_{30} = \frac{193785856000000}{651}, R_{36} = \frac{36150983979827200000000}{191919}$, and so on.

It can be seen that the sequence R_{6n} grows rather quickly, so we may hope to find an asymptotic formula for R_{6n}. We can do so by noticing that $G_{6k}(\omega) \sim 6$. This can be seen by recalling the definition of G_{6k} as $G_{6k}(\omega) = \sum_{(m,n)\in\mathbb{Z}^2/(0,0)} \frac{1}{(m+n\omega)^{6k}}$. We can separate the sum into 2 different sums: one sum contains only the values of (m,n) such that $|m+n\omega| \leq 1$ and the other sum contains values of (m,n) such that $|m+n\omega| > 1$. The first sum has only finitely many terms and summing

over each of them individually gives a value of 6. The second sum vanishes for $k \to \infty$ since $|m + n\omega| > 1$. However, $G_{6k}(\omega) = \frac{R_{6k}\pi_3^{6k}}{(6k)!} \sim 6$. Thus, $R_{6n} = \frac{6(6n)!}{\pi_3^{6n}}(1 + O(1))$. Using Stirling's formula then gives us the following asymptotic formula:

20.
$$R_{6n} \sim 12\sqrt{3\pi n}\left(\frac{6n}{e\pi_3}\right)^{6n}$$

for $n \to \infty$.